



DIGITAL VLSI DESIGN

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DIGITAL VLSI DESIGN

PART 1

MOS Inverters (Static Characteristics)

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DIGITAL VLSI DESIGN

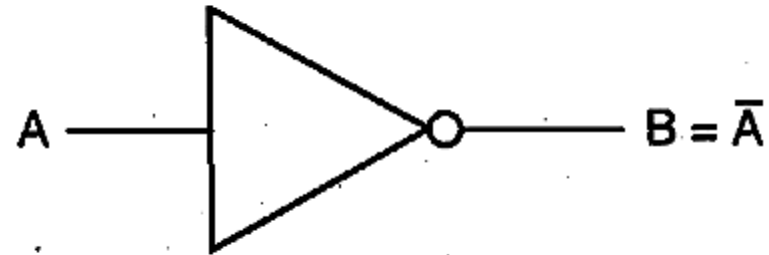
MOS Inverters (Static Characteristics)

Part b – Introduction to Inverters

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MOS Inverters (Static Characteristics)



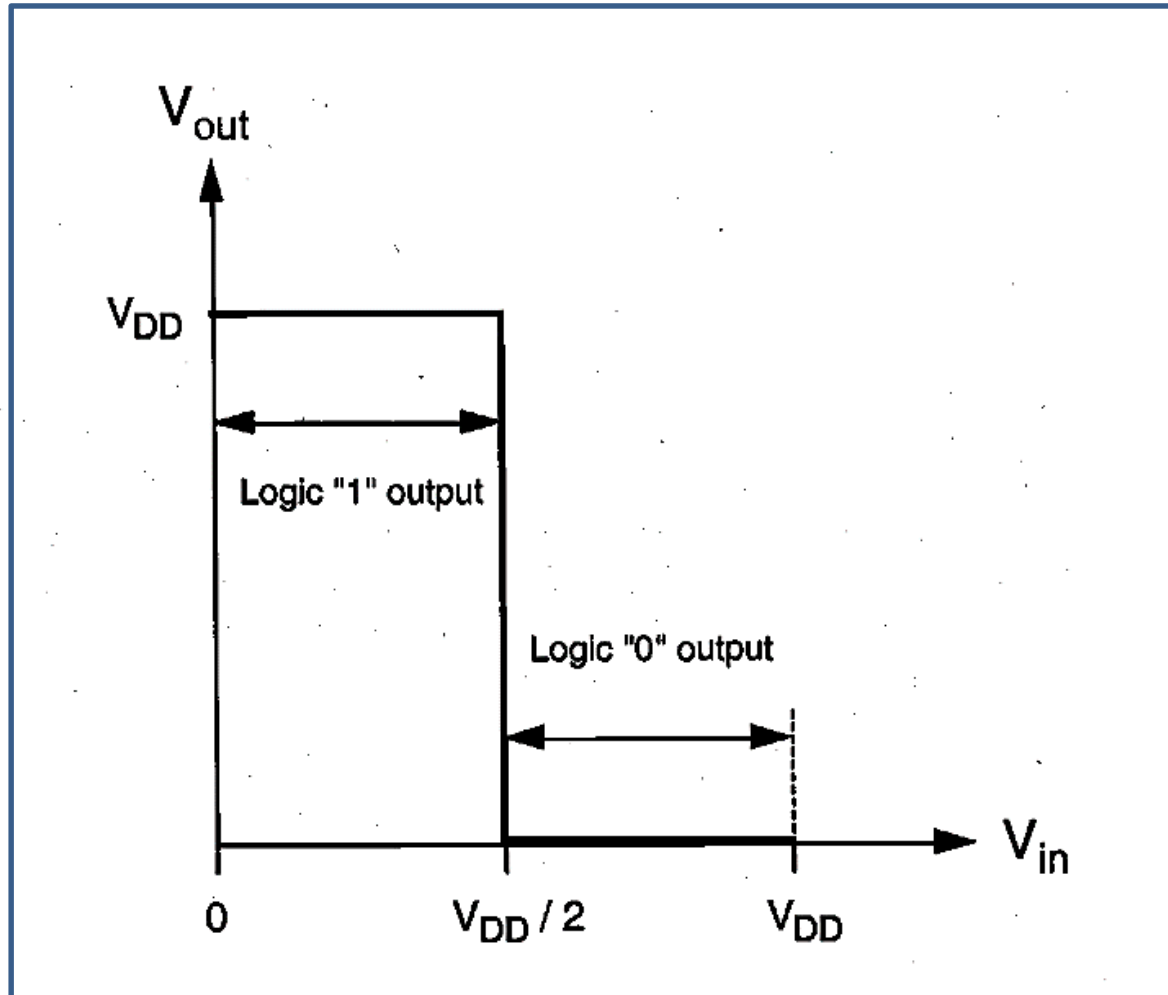
Symbol

A	B
0	1
1	0

Truth Table

Logic symbol and truth table of the inverter

MOS Inverters (Static Characteristics)



For an Ideal Inverter

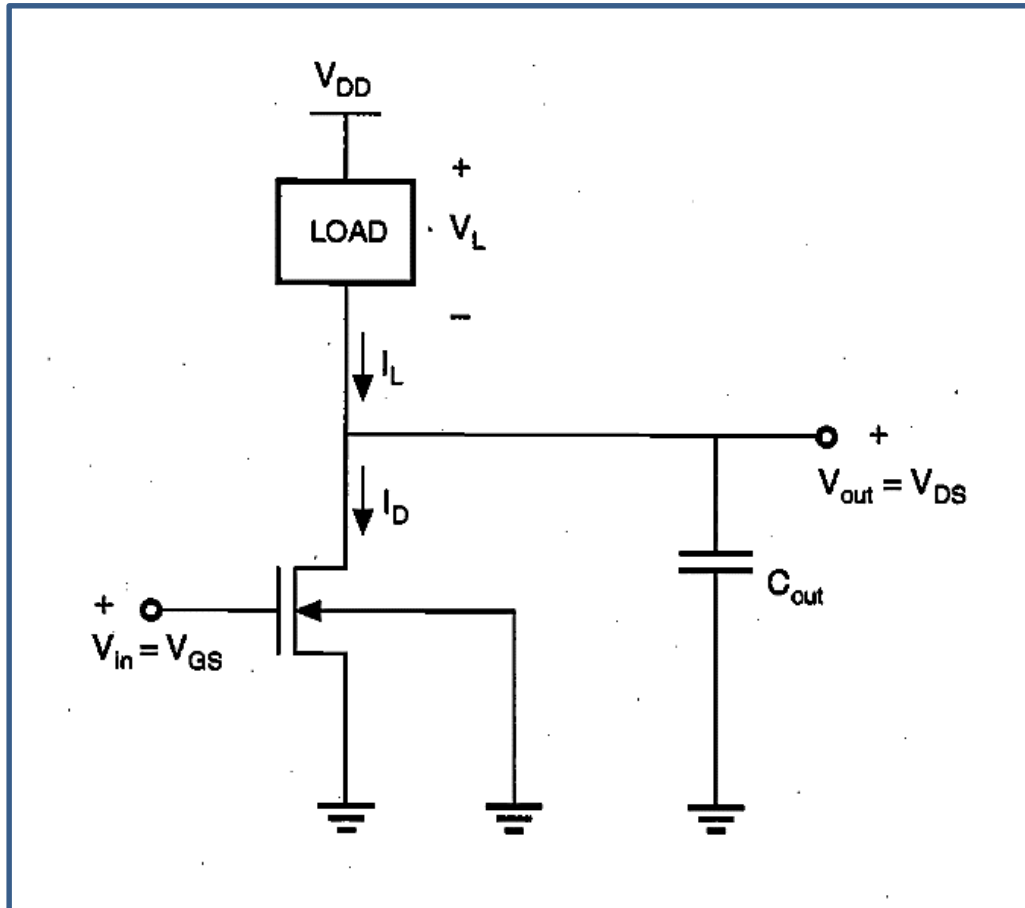
- An input voltage $0 < V_{in} < V_{th}$ is interpreted by this ideal inverter as a logic "0"
- An input voltage $V_{th} < V_{in} < V_{DD}$ is interpreted as a logic "1"

Where V_{th} is called the inverter threshold voltage

Voltage transfer characteristic (VTC) of the ideal inverter

MOS Inverters (Static Characteristics)

The generalized circuit structure of an nMOS inverter



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MOS Inverters (Static Characteristics)

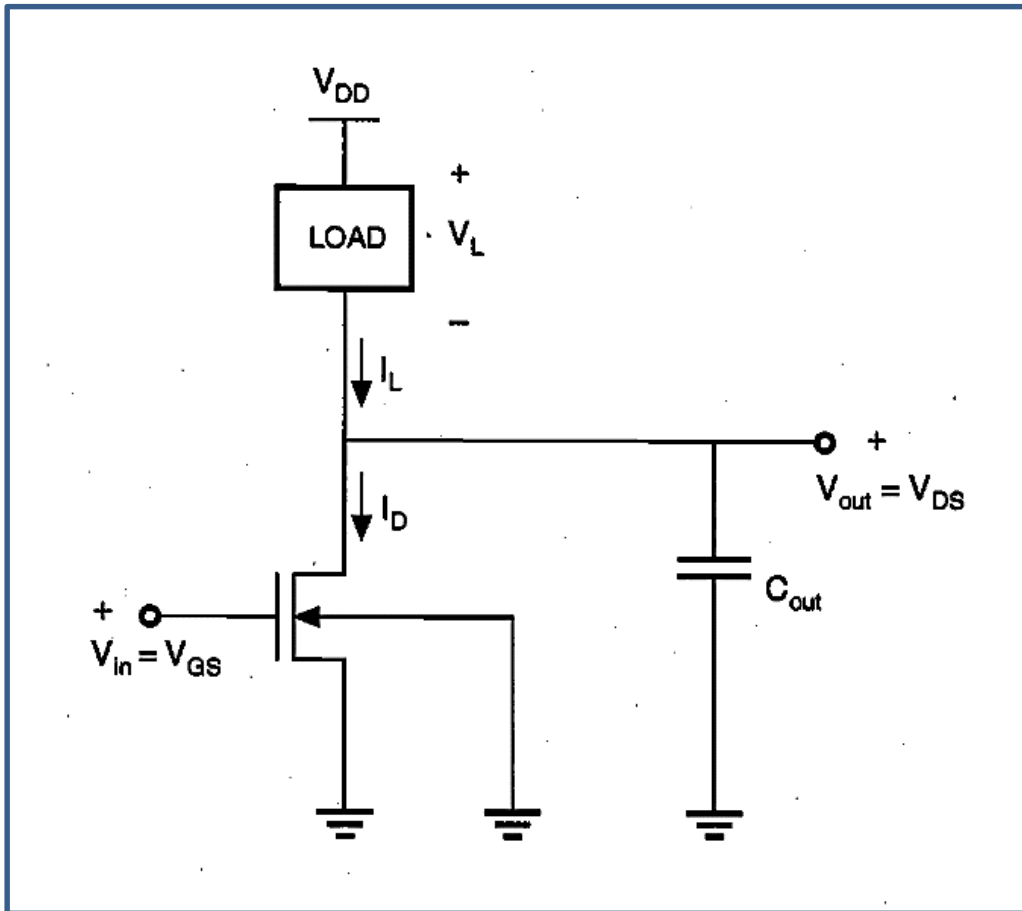
5 Critical Voltages – Inverter

Vinay Reddy Narayana

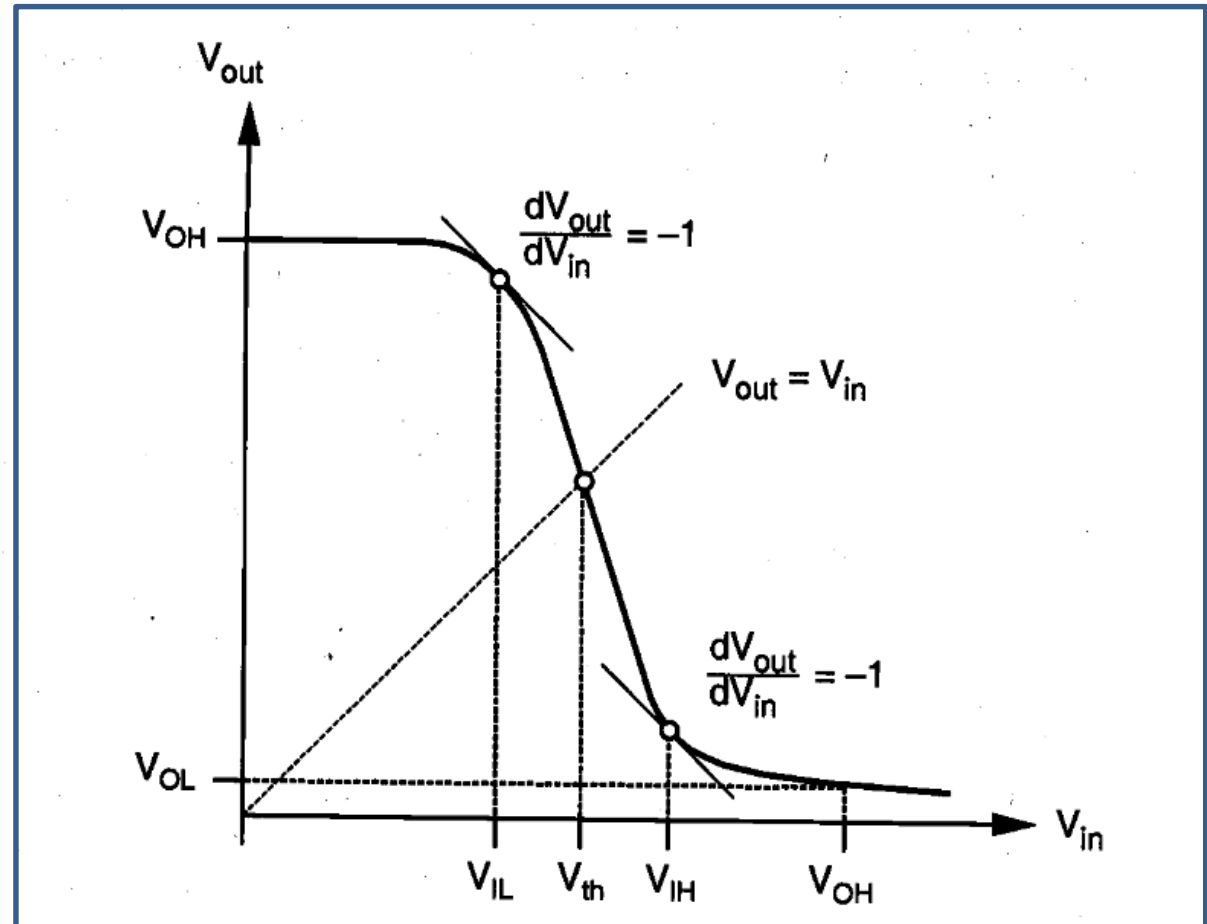
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MOS Inverters (Static Characteristics)

The generalized circuit structure of an nMOS inverter

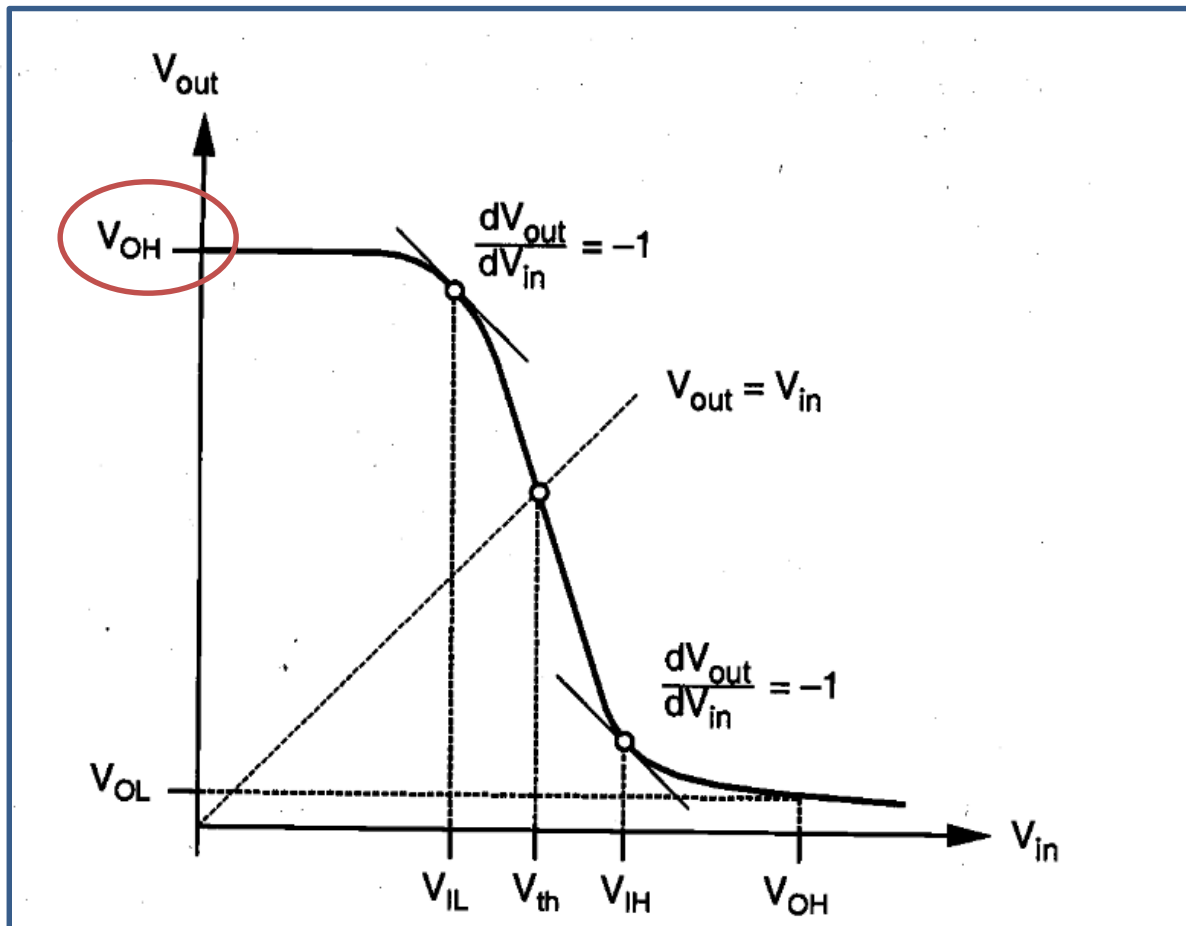


Typical voltage transfer characteristic (VTC) of a realistic nMOS inverter.



MOS Inverters (Static Characteristics)

Typical voltage transfer characteristic (VTC) of a realistic nMOS inverter.



CONCLUSIONS

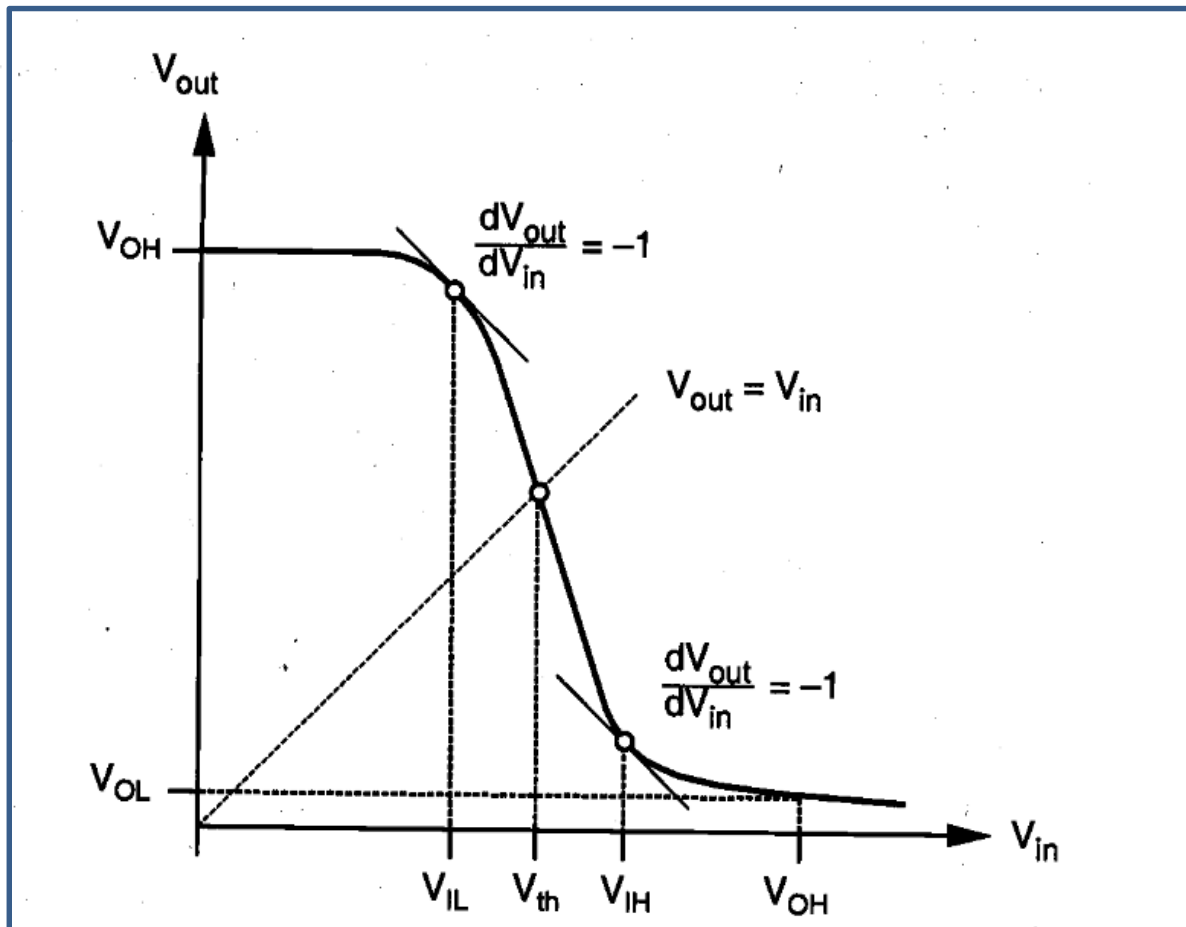
For very low input voltage levels, the output voltage V is equal to the high value of **VOH (output high voltage)**.

The driver nMOS transistor is in cut-off, and hence, does not conduct any current.

Consequently, the voltage drop across the load device is very small in magnitude, and the output voltage level is high.

MOS Inverters (Static Characteristics)

Typical voltage transfer characteristic (VTC) of a realistic nMOS inverter.



CONCLUSIONS

We identify two critical voltage points on this curve, where the slope of the $V_{out}(V_{in})$ characteristic becomes equal to -1

$$\frac{dV_{out}}{dV_{in}} = -1$$

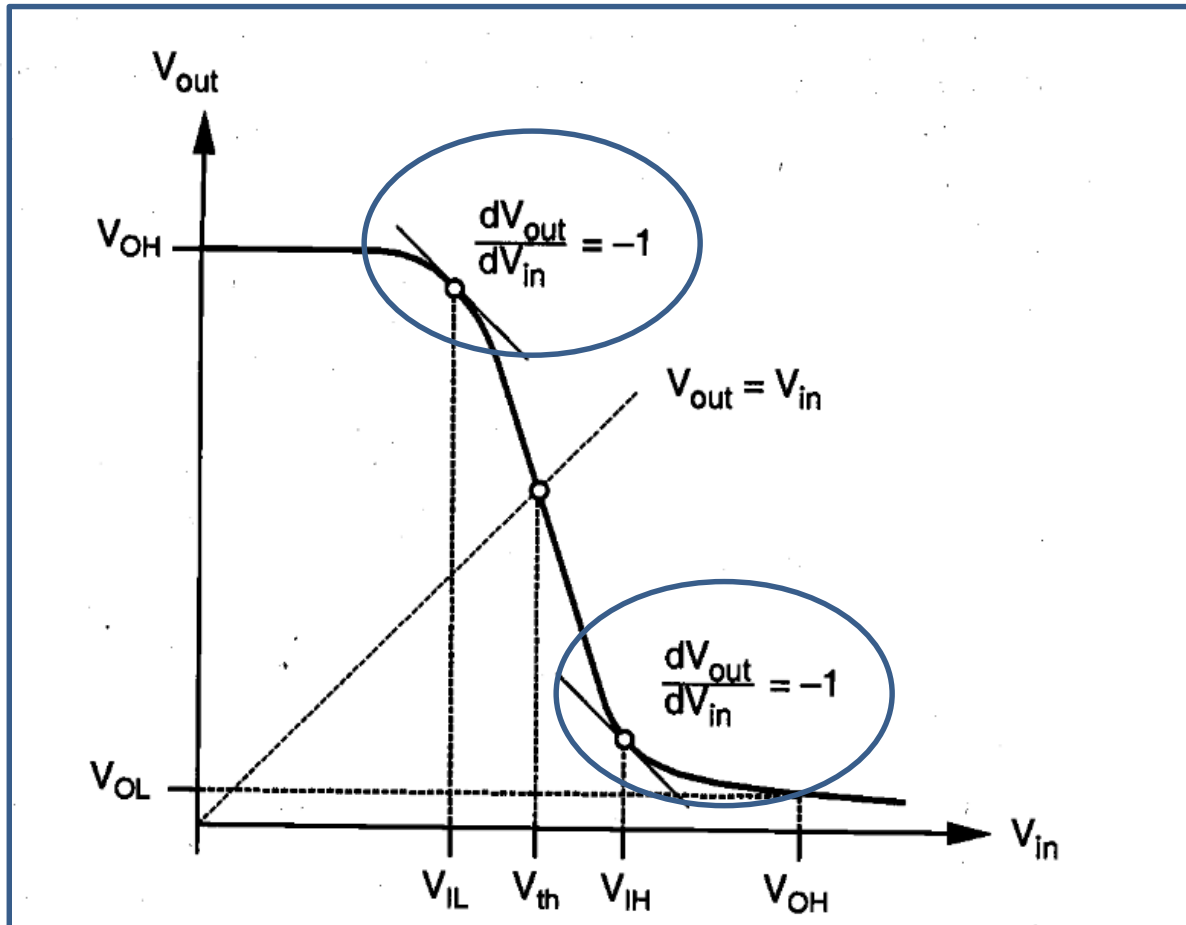
Meaning:

$V_{out} = -V_{in}$

Output and Input are inverted

MOS Inverters (Static Characteristics)

Typical voltage transfer characteristic (VTC) of a realistic nMOS inverter.



CONCLUSIONS

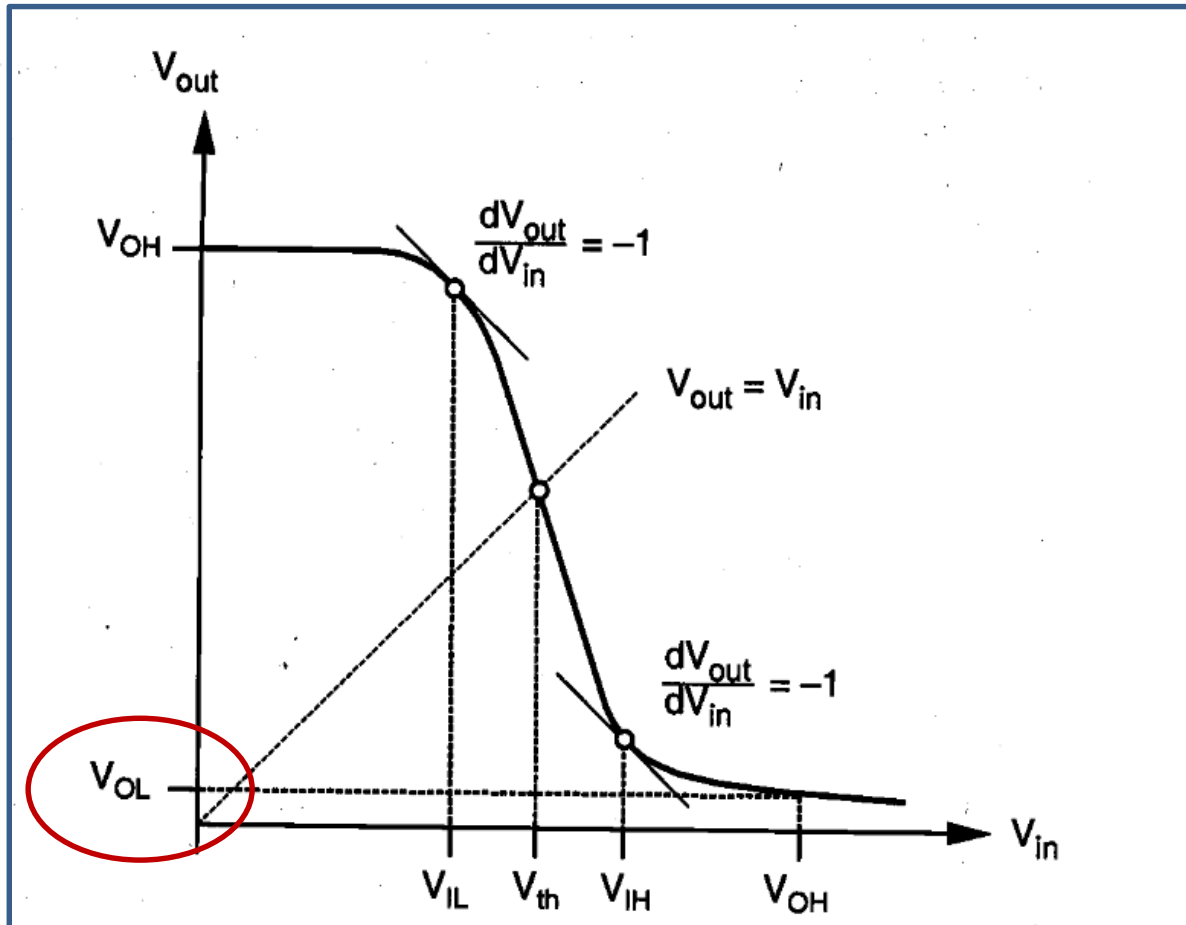
The smaller input voltage value satisfying this condition is called the **input low voltage V_{IL}** .

The larger input voltage satisfying this condition is called the **input high voltage V_{IH}** .

*Note: Both of these voltages play significant roles in determining the **noise margins** of the inverter circuit*

MOS Inverters (Static Characteristics)

Typical voltage transfer characteristic (VTC) of a realistic nMOS inverter.

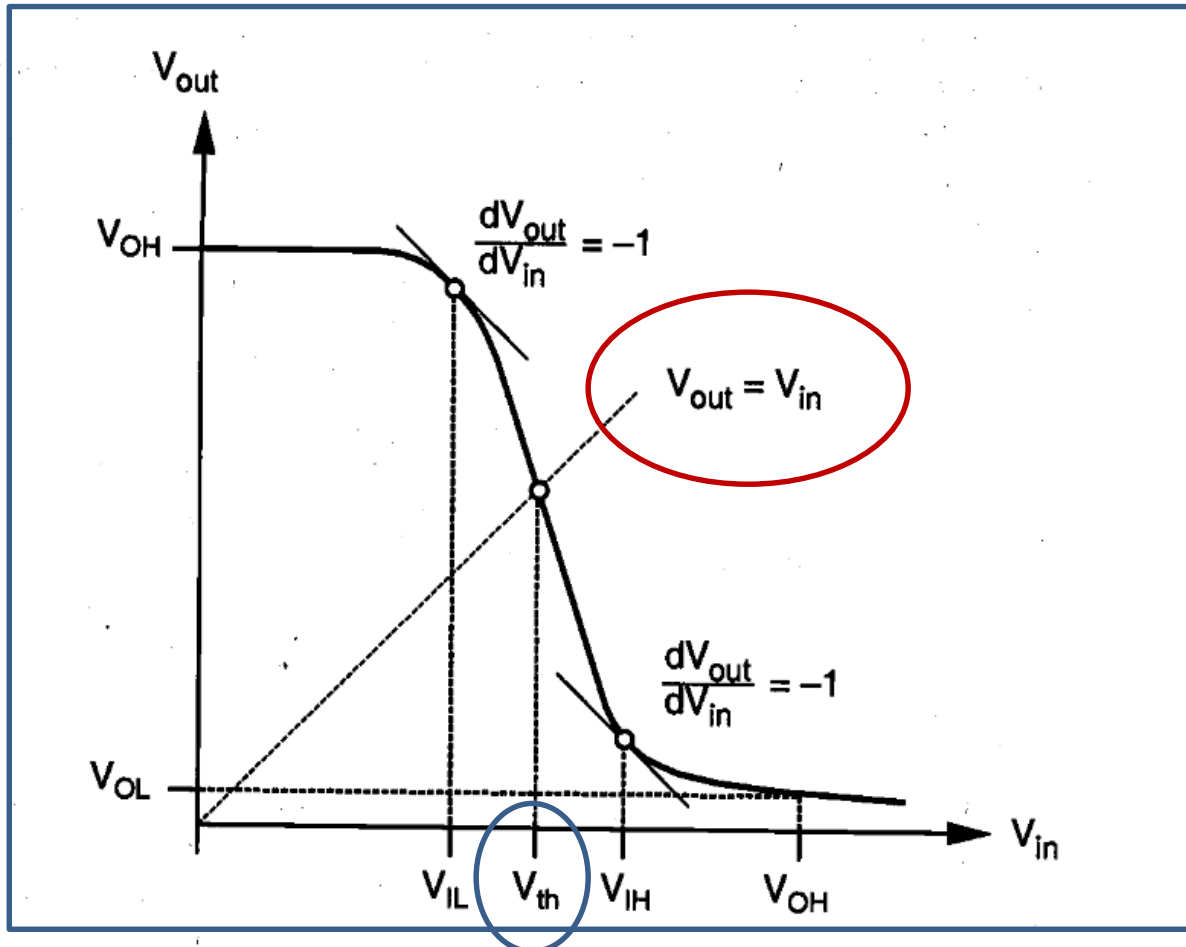


CONCLUSIONS

As the input voltage is further increased, the output voltage continues to drop and reaches a value of **VOL (output low voltage)** when the input voltage is equal to V_{OH} .

MOS Inverters (Static Characteristics)

Typical voltage transfer characteristic (VTC) of a realistic nMOS inverter.

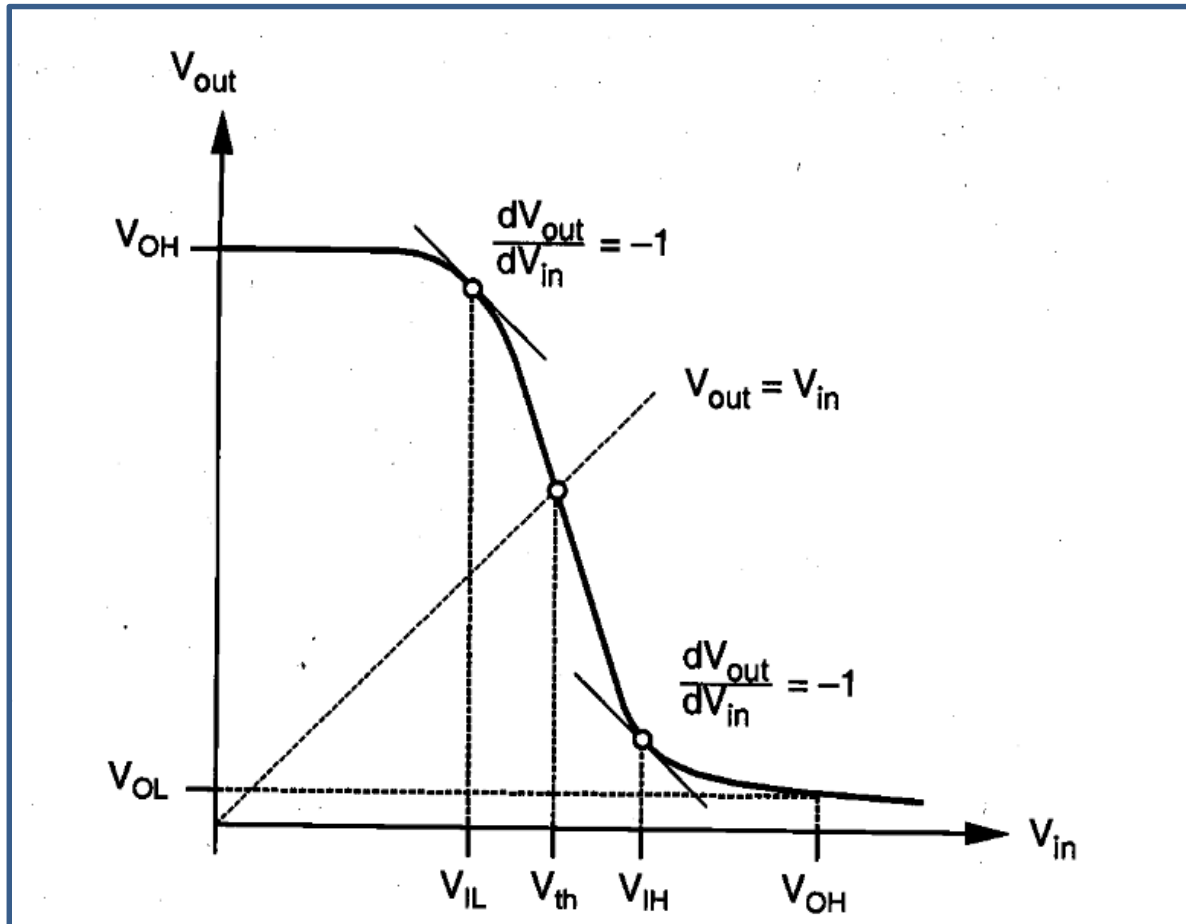


CONCLUSIONS

The **inverter threshold voltage V_{th}** , which is considered as the transition voltage, is defined as the point where $V_{in} = V_{out}$, on the VTC

MOS Inverters (Static Characteristics)

Typical voltage transfer characteristic (VTC) of a realistic nMOS inverter.



CONCLUSIONS

Thus, a total of five critical voltages, **VOL**, **VOH**, **VIL**, **VIH**, and **Vth**, characterize the DC input-output voltage behavior of the inverter circuit.

- VOH** : Minimum output voltage when the output level is logic "1".
- VOL** : Maximum output voltage when the output level is logic "0".
- VIL** : Maximum input voltage which can be interpreted as logic "0".
- VIH** : Minimum input voltage which can be interpreted as logic "1".

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MOS Inverters (Static Characteristics)

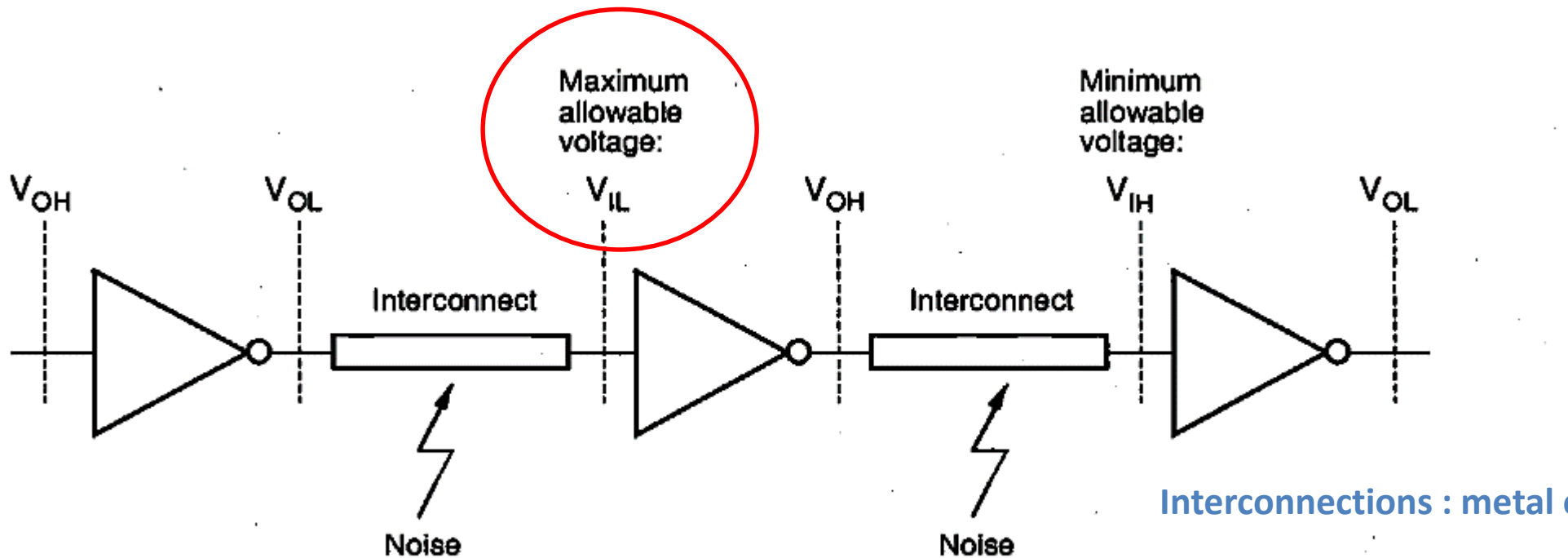
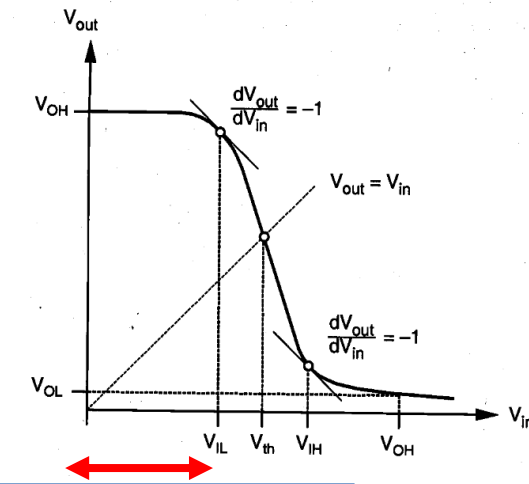
Noise Margin - NM

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MOS Inverters (Static Characteristics)

Propagation of digital signals under the influence of noise

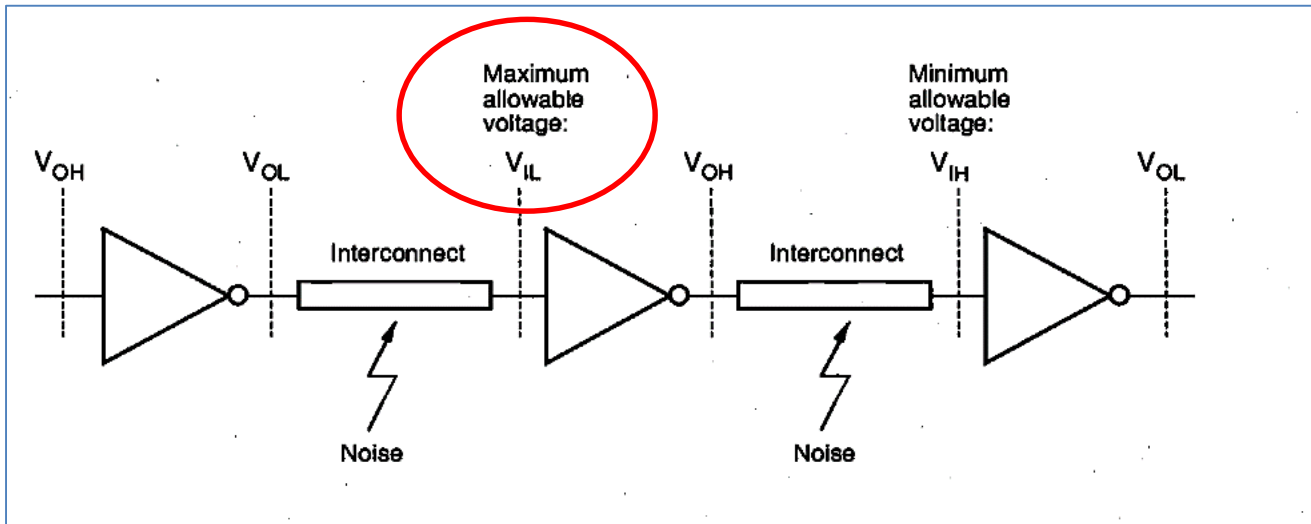


Interconnections : metal or polysilicon line

Since on-chip interconnects are generally prone to signal noise, the output signal of the first inverter will be perturbed during transmission.

MOS Inverters (Static Characteristics)

Propagation of digital signals under the influence of noise



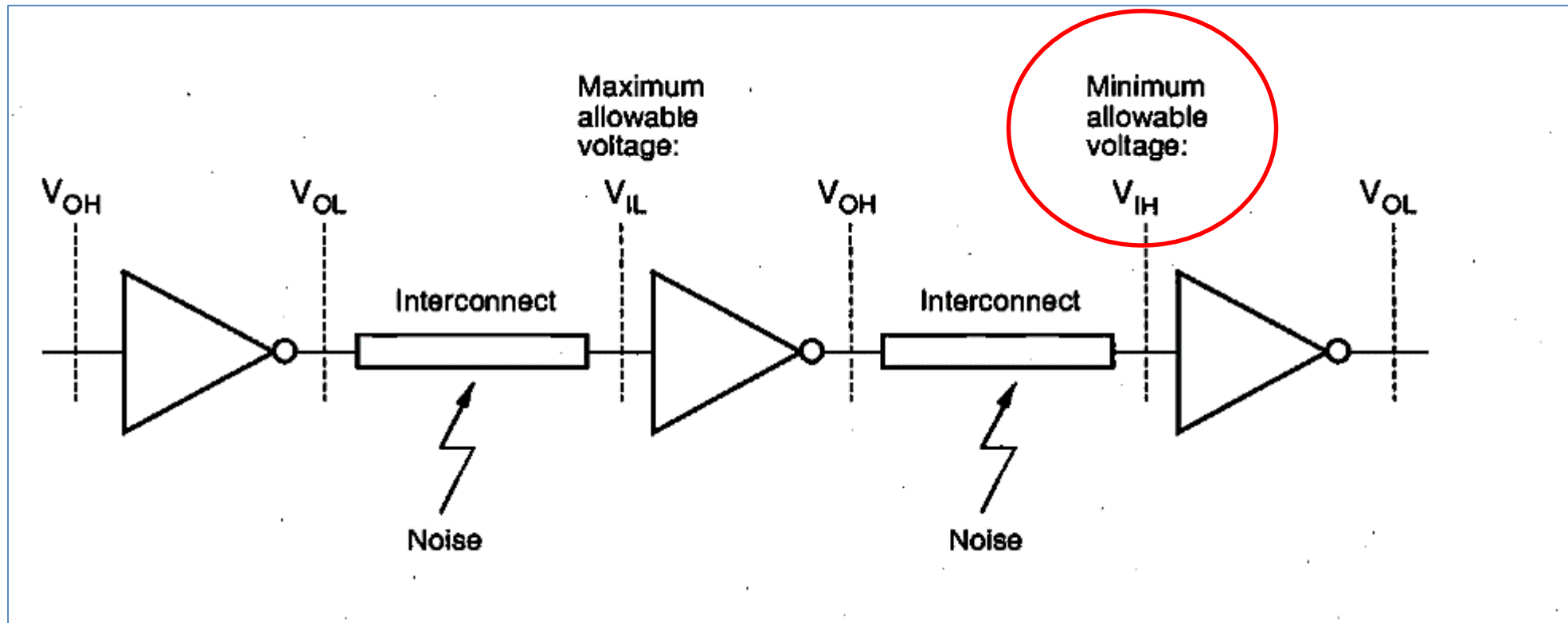
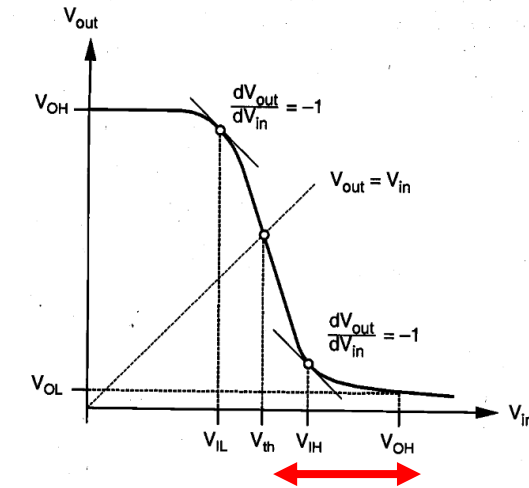
The voltage level at the input of the second inverter will be either larger or smaller than V_{OL} .

Case1: If the input voltage of the second inverter is smaller than V_{OL} this signal will be interpreted correctly as a logic "0" input by the second inverter.

Case2: On the other hand, if the input voltage becomes larger than V_{IL} as a result of noise, then it may not be interpreted correctly by the inverter.

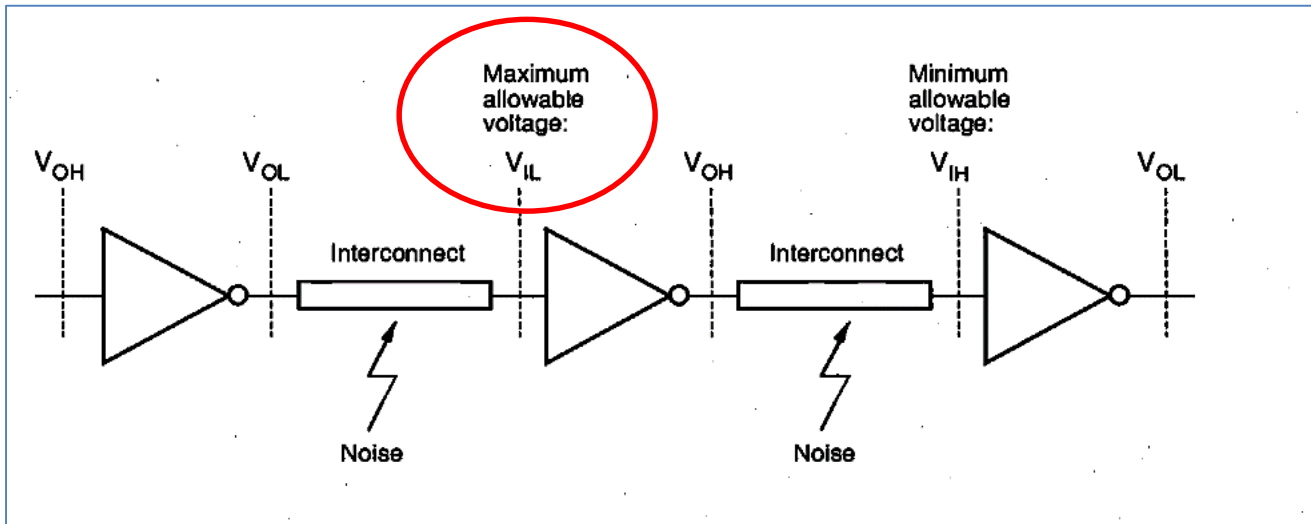
MOS Inverters (Static Characteristics)

Propagation of digital signals under the influence of noise



MOS Inverters (Static Characteristics)

Propagation of digital signals under the influence of noise



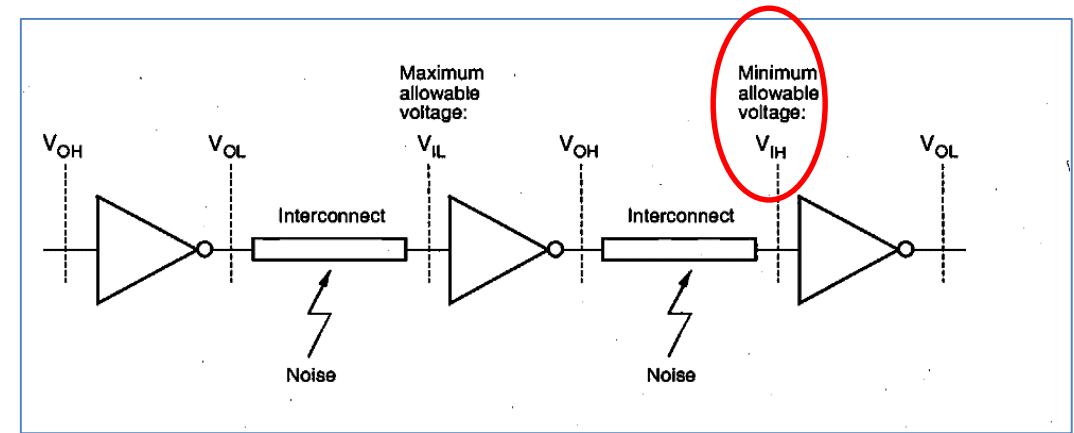
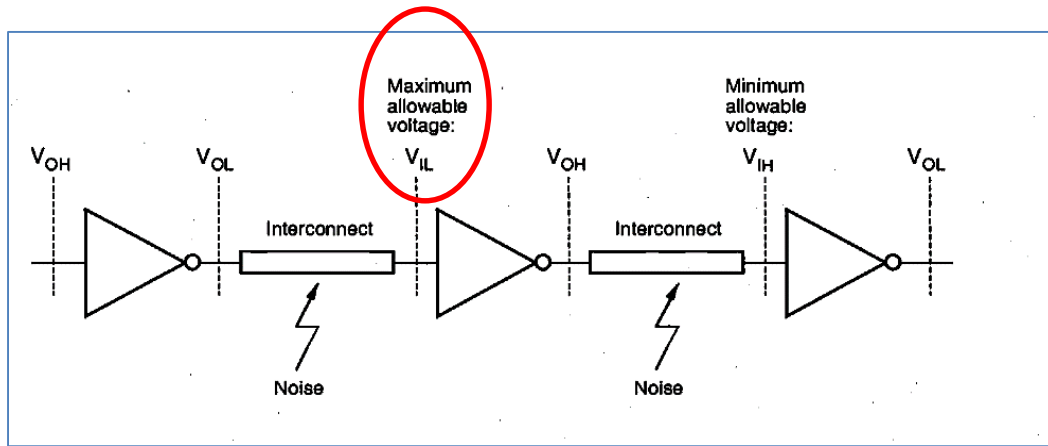
The voltage level at the input of the third inverter will be either larger or smaller than V_{OH} .

Case1: If the input voltage of the third inverter is larger than V_{OH} this signal will be interpreted correctly as a logic "1" input by the third inverter.

Case2: On the other hand, if the input voltage becomes smaller than V_{IH} as a result of noise, then it may not be interpreted correctly by the inverter.

MOS Inverters (Static Characteristics)

Propagation of digital signals under the influence of noise



CONCLUSIONS

Thus, we conclude that V_{IL} is the maximum allowable voltage at the input of the second inverter, which is low enough to ensure a logic "1" output.

Consequently, V_{IH} is the minimum allowable voltage at the input of the third inverter which is high enough to ensure a logic "0" output.

NOISE MARGIN:

These observations lead us to the definition of noise tolerances for digital circuits called noise margins and denoted by NM.

The noise immunity of the circuit increases with NM.

Two noise margins will be defined:

the noise margin for low signal levels (NML) and
the noise margin for high signal levels (NMH).

$$NM_L = V_{IL} - V_{OL}$$

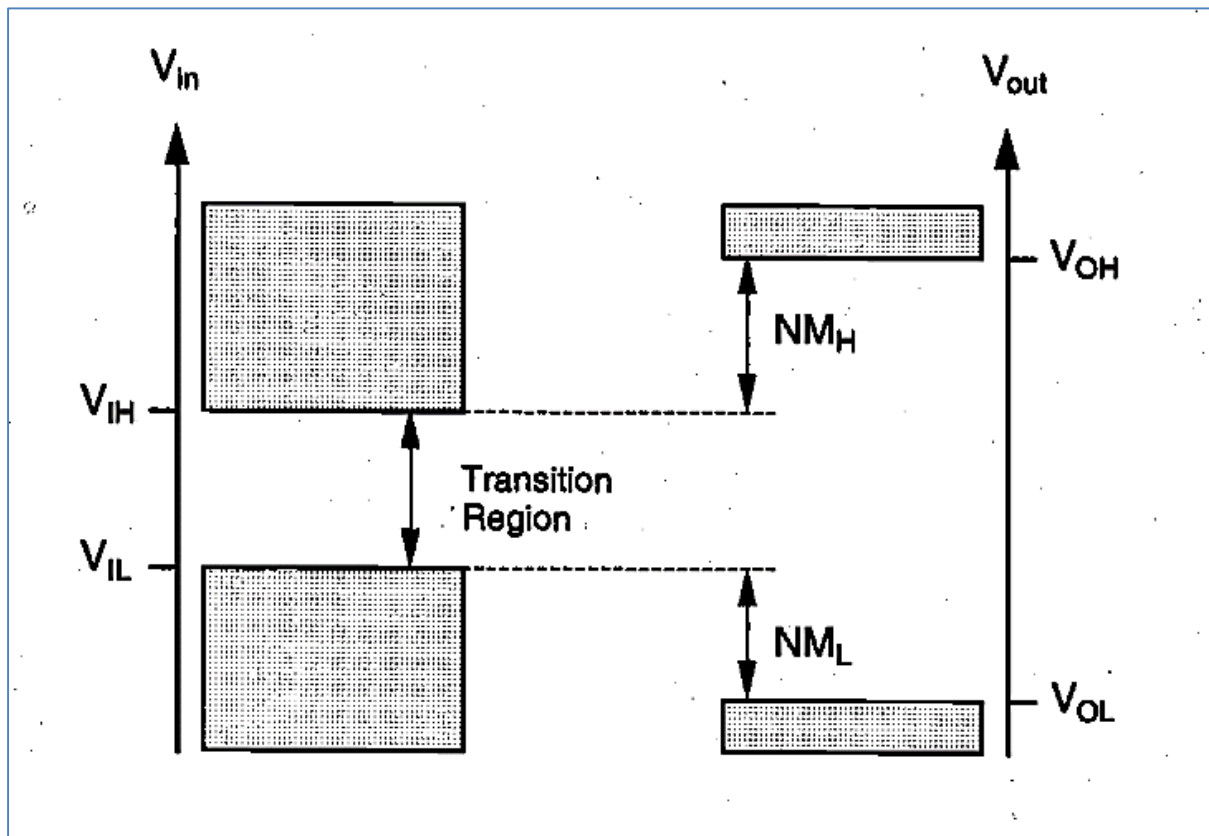
$$NM_H = V_{OH} - V_{IH}$$

MOS Inverters (Static Characteristics)

VLSI Design Flow (ASIC Design Flow)

Definition of noise margins NML and NMH.

Note that the shaded regions indicate valid high and low levels for the input and output signals.



The key idea

For OUTPUTS:

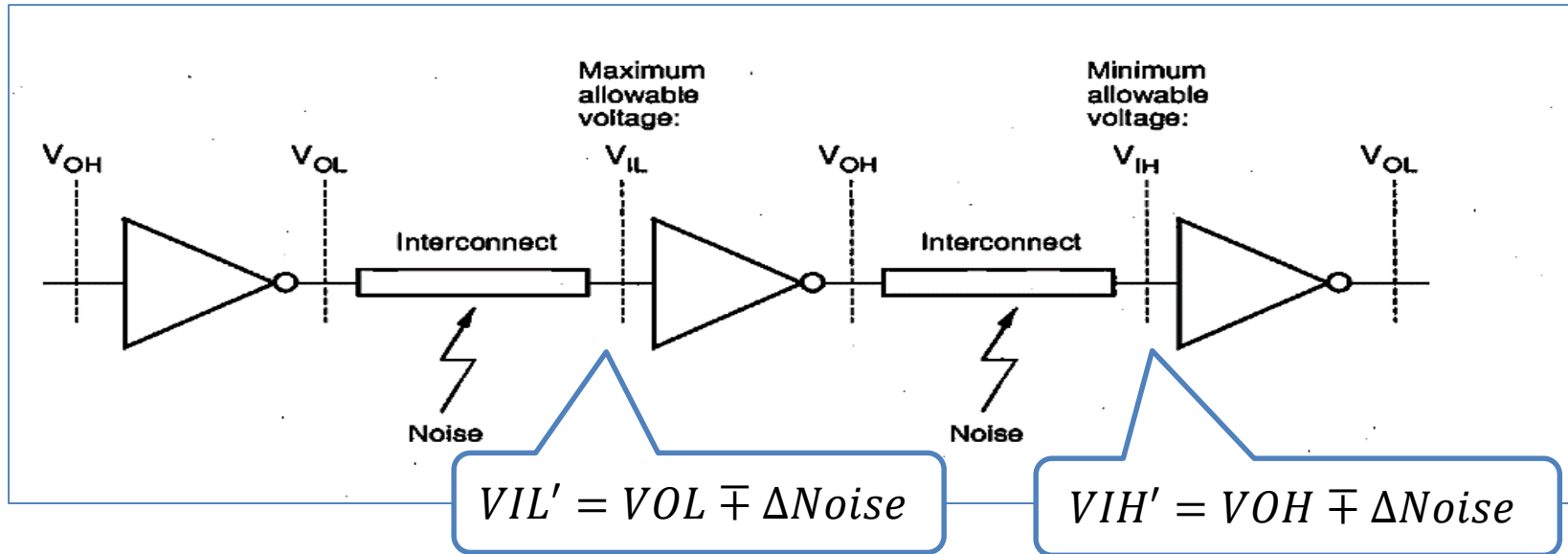
Actual output HIGH can go up to V_{DD} , but we only guarantee $\geq V_{OH}(\min)$

Actual output LOW can go down to 0, but we only guarantee $\leq V_{OL}(\max)$

Note: Voltage range between V_{IL} and V_{IH} corresponds to V_{in} that may not be processed correctly as Logic 0 or logic 1 by inverter, This region is called Uncertain or Transition region

MOS Inverters (Static Characteristics)

Propagation of digital signals under the influence of noise



Output of inverter under noise free conditions

$$V_{out} = f(v_{in}) \text{ -----(1)}$$

MOS Inverters (Static Characteristics)

V_{in} is unsettled from the nominal values because of external influence (ΔV_{noise})
Therefore output voltage will deviate from nominal value

$$V_{out}' = f(V_{in} + \Delta V_{noise}) \text{ -----(2)}$$

By using simple first-order Taylor series expansion and neglecting Higher order terms,

$$V_{out}' = f(V_{in}) + \frac{dV_{out}}{dV_{in}} \cdot \Delta V_{noise} + \text{Higher order terms) ----- (3)}$$

Unsettled output = Nominal o/p + Gain x External Influence

MOS Inverters (Static Characteristics)

$$V_{out}' = f(V_{in}) + \frac{dV_{out}}{dV_{in}} \cdot \Delta V_{noise} + \text{Higher order terms) ----- (3)}$$

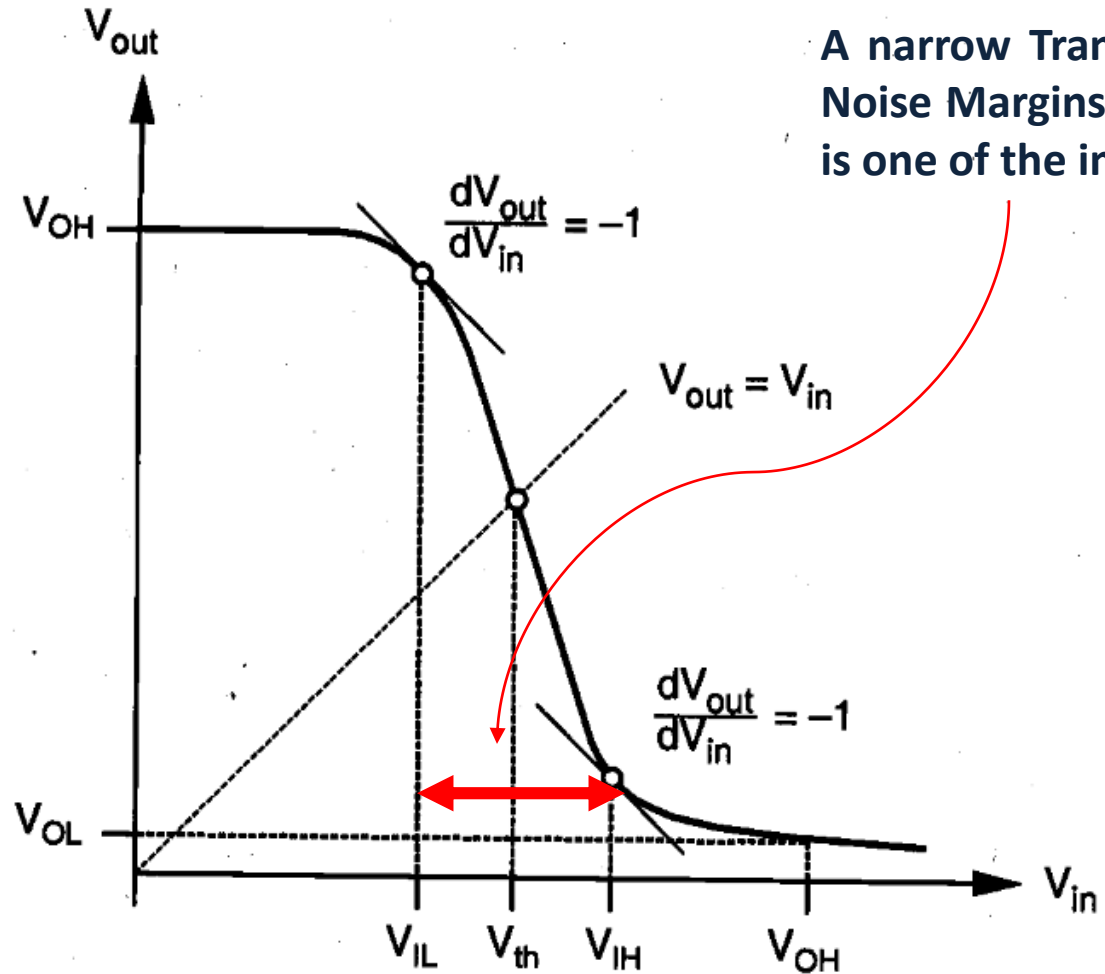
Unsettled output = Nominal o/p + Gain x External Influence

CONCLUSIONS

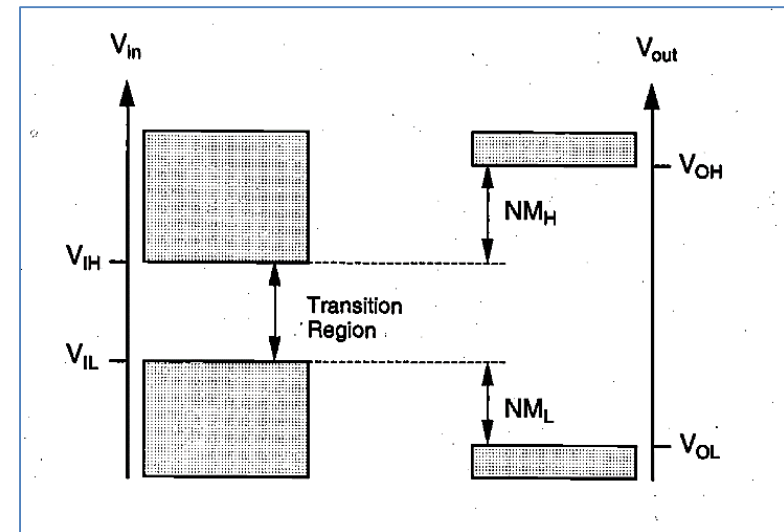
If the magnitude of the voltage gain at the nominal input voltage V_{in} is smaller than unity, then the input perturbation is not amplified and, consequently, the output perturbation remains relatively small. Otherwise, with the voltage gain larger than unity, a small perturbation in the input voltage level will cause a rather large perturbation in the output

Hence boundaries of valid input signal regions as voltage points where the magnitude of inverter voltage gain is equal unity.

MOS Inverters (Static Characteristics)



A narrow Transition (Uncertain) region obviously allow large Noise Margins. Therefore reducing the transition region width is one of the important design objective.



$$NM_L = V_{IL} - V_{OL}$$

$$NM_H = V_{OH} - V_{IH}$$

Power and Area Considerations

Power : $P_{DC} = V_{DD} \cdot I_{DC}$

If the duty cycle of input pulse is 50% th

$$P_{DC} = \frac{V_{DD}}{2} [I_{DC} (V_{in} = Low) + I_{DC} (V_{in} = High)]$$

Area : W/L should be near unity in order to optimize the area but as it contradicts the other design criteria like noise margin , dynamic switching speed and driving capability... there will be a trade off



THANK YOU

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